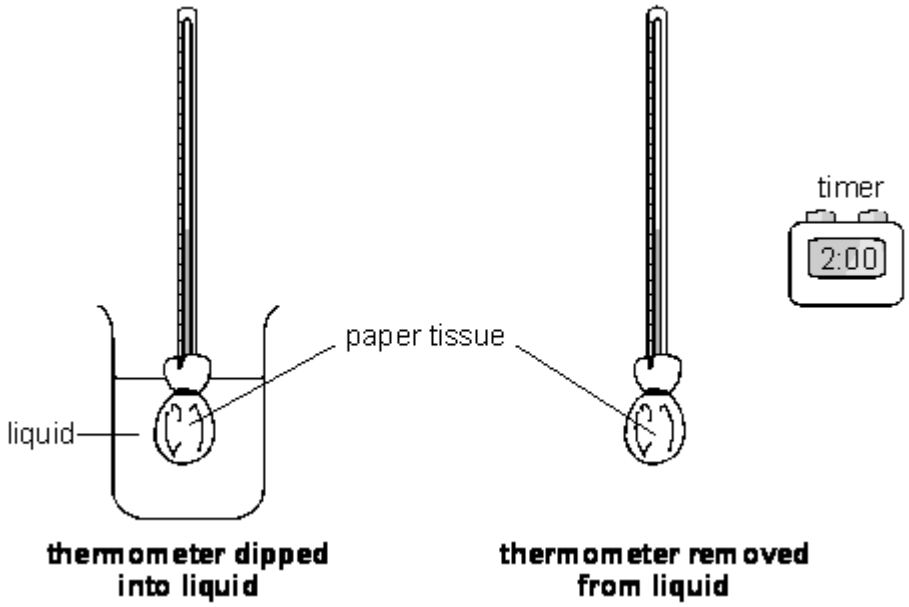
Q1.

A group of pupils carried out an experiment with four different liquids.

They wrapped paper tissues around the bulbs of four thermometers. They secured the tissues with rubber bands. Each thermometer was then dipped into a different liquid and removed. The temperature was recorded. The reading on each thermometer was then noted every two minutes.



The results are given in the table below.

time in min	reading, in °C, on the thermometer dipped in:			
	propanone	**ethanol**	**ether**	**water**
0	23	23	23	23
2	13	19	11	21
4	4	16	−2	20
6	2	14	−8	20
8	3	14	−1	20
10	4	14	8	20
12	6	14	17	20

(a) Suggest which liquid evaporated most rapidly.

.....

1 mark

(b) After six minutes, the reading went up on the thermometer dipped in ether. Explain why.

.....

.....

1 mark

(c) What is the likely temperature of the room in which they did the experiment?

.....

1 mark

(d) The pupils carried out the experiment with propanone again. However, this time they did **not** wrap the thermometer in paper tissue but just dipped the glass bulb into the propanone and removed it. Suggest how their second set of results would be different.

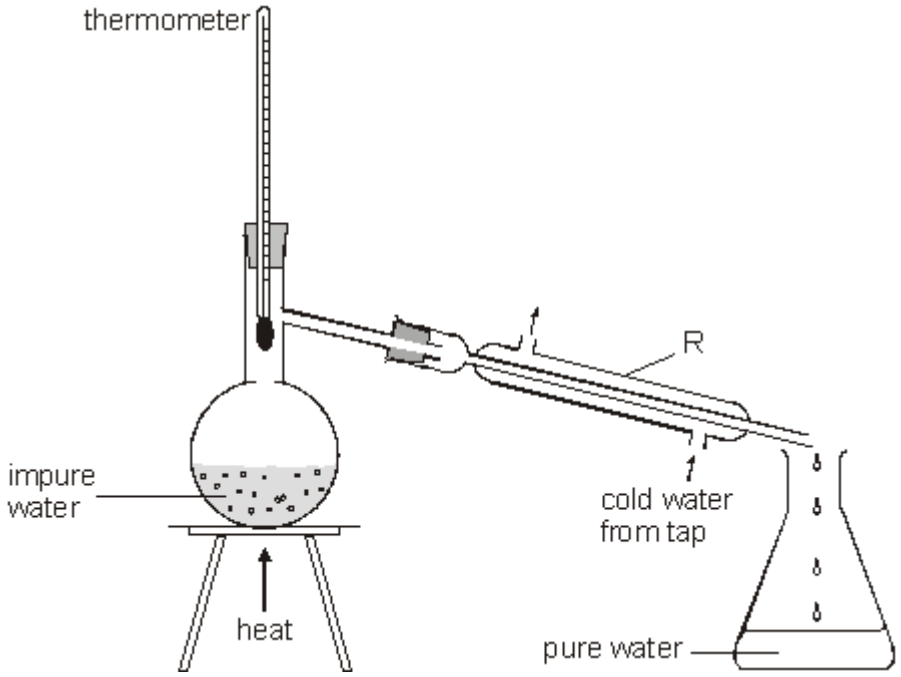
.....

.....

1 mark
Maximum 4 marks

Q2.

(a) The apparatus in the diagram below is used to obtain pure water from impure water.



(i) What temperature would the thermometer show?

.....°C

1 mark

(ii) What is the function of the piece of apparatus labelled R?

.....

.....

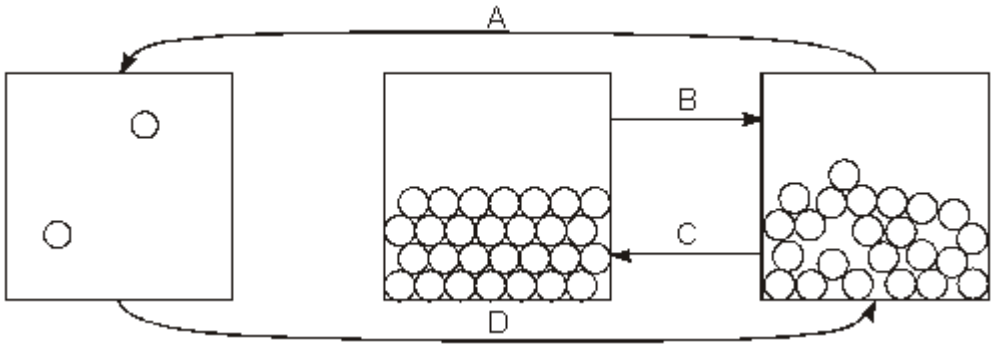
1 mark

(iii) Give the name of the process which purifies water in this way.

.....

1 mark

(b) The diagram below shows particles in a gas, a solid and a liquid.
Each arrow, A, B, C and D, represents a change of state.



(i) Choose from the following words to complete the sentences below.

boiling **condensing** **distilling** **evaporating**

filtering **freezing** **melting**

Change of state A is called ..... .

Change of state B is called ..... .

Change of state C is called ..... .

Change of state D is called ..... .

4 marks

(ii) Look back to the apparatus in part (a).
Give the letter, A, B, C or D, from the diagram above, for the change of state which occurs:

in the round-bottomed flask .....

in the piece of apparatus labelled R. .....

2 marks
Maximum 9 marks

Q1.

(a) ether

1

(b) all the ether had evaporated

accept 'all the ether had gone'
*do **not** accept 'it was warmed by the room'*

1

(c) 23°C

*do **not** accept '23'*

1

(d) any **one** from the

- temperatures would not go as low

  accept 'the temperatures would be higher'
  *do **not** accept 'the temperatures would go higher'*

- the temperatures would go up again much sooner

  accept 'the readings would have to be taken more frequently'
  *do **not** accept 'the temperatures would go down more quickly'*
  ***or** 'it would evaporate more quickly'*

1

[4]

Q2.

(a) (i) 100

accept answers from 98 to 100

1 (L5)

(ii) to condense the water vapour

accept 'to change the gas into liquid'
***or** 'condensation' or 'condenser'*
accept 'to cool the vapour into water'
*do **not** accept 'to cool the vapour or water'*

1 (L5)

(iii) distillation **or** distilling

1 (L5)

(b) (i) evaporating **or** boiling

1 (L6)

melting

1 (L6)

freezing

1 (L6)

condensing

1 (L6)

(ii) A

accept 'boiling'

1 (L6)

D

accept 'condensing'

accept the letters written in the correct places on the diagram

1 (L6)

[9]